

Development of an IoT-Based Smart Sub-Metering System for Room-Level Electricity Monitoring, Carbon Emission Estimation, and Fair Electricity Cost Allocation

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I. INTRODUCTION

Shared residential spaces often rely on a common electricity meter and divide costs equally among occupants, even when their actual electricity consumption differs. Occupants may use different appliances, follow different daily routines, and have different economic circumstances. As a result, equal bill sharing may not accurately reflect individual electricity use or the financial burden of electricity consumption.

IoT-based smart sub-metering provides a practical approach for measuring electricity consumption at the individual room level. Previous studies have shown substantial differences in dormitory electricity consumption due to occupant behavior, room characteristics, and occupancy patterns [1], [3], [5]. IoT-based systems have also demonstrated the feasibility of continuous, high-resolution electricity monitoring and real-time visualization [2], [6], [8]. However, existing studies have generally focused on electricity monitoring, energy management, or system development separately, with less attention to combining room-level consumption measurement with carbon-emission estimation and electricity-cost allocation.

This study develops an IoT-based smart sub-metering system integrated with a web-based monitoring platform to address this gap. The system measures room-level electricity consumption and uses these measurements to estimate carbon emissions and electricity costs. In addition, economic position is considered as an additional factor when evaluating electricity-cost fairness. The study involves two working occupants with different economic positions: a journalist residing in Room K and a medicine company representative residing in Room F with a comparatively lower economic position.

The study therefore examines differences in electricity consumption, carbon emissions, and electricity cost burden between the two occupants and considers how these differences relate to their economic circumstances. The developed web-based platform provides a practical interface for monitoring, visualizing, and analyzing the collected electricity data.

The main contributions of this study are:

- Development of an IoT-based room-level electricity monitoring system integrated with a web-based platform.
- Estimation of individual carbon emissions from measured electricity consumption.
- Comparison of conventional equal-bill and consumption-based electricity cost allocation.
- Inclusion of economic position as an additional factor when evaluating electricity-cost fairness.
- Practical evaluation using 15 days of real electricity measurements from two working occupants with different economic positions.

A. Background

Electricity consumption in shared residential environments varies between occupants because of differences in appliance use, daily routines, and occupancy patterns. Previous dormitory studies have found substantial differences in room-level electricity consumption and identified factors such as appliance use, room location, and academic schedules as important influences [1], [3], [5]. These findings support the need for more detailed monitoring than a single shared electricity meter can provide.

IoT-based smart metering can address this problem by providing continuous electricity measurements at the room or equipment level. Studies have demonstrated the feasibility of collecting high-resolution electricity data and using IoT systems for real-time monitoring and energy management [2], [6], [8]. Smart monitoring platforms have also been shown to improve energy awareness and support energy-management decisions [4], [7].

However, conventional shared billing generally records only the total electricity consumption of a residence. It does not show how much electricity each occupant used, the carbon emissions associated with that use, or whether each occupant's share of the bill reflects their actual consumption. Room-

level monitoring can address these issues by linking individual electricity use with cost and environmental impact.

Fairness may also depend on economic circumstances. Two occupants paying the same amount may experience different financial burdens if their economic positions differ. Therefore, considering economic position alongside measured electricity consumption provides an additional perspective when evaluating electricity-cost fairness.

This study applies IoT-based sub-metering to two working occupants with different economic positions and uses a web-based platform to monitor their electricity use. The system measures consumption, estimates carbon emissions, and compares equal-bill with consumption-based cost allocation. In this way, the study extends previous work on electricity monitoring by combining consumption, environmental impact, and cost allocation at the room level.

B. Research Objectives

The aim of this study is to develop and evaluate an IoT-based smart sub-metering system for individual electricity monitoring, carbon emission estimation, and fair electricity cost allocation while considering occupants' economic positions. The specific objectives are:

- 1) To develop an IoT-based smart sub-metering system for room-level electricity monitoring.
- 2) To measure and analyze individual electricity consumption using the developed system.
- 3) To estimate individual carbon emissions from measured electricity consumption.
- 4) To compare equal-bill and consumption-based electricity cost allocation.
- 5) To examine the relationship between economic position, electricity consumption, carbon emissions, and cost burden.

C. Research Questions

The main research question addressed in this study is: how can an IoT-based smart sub-metering system support individual electricity monitoring, carbon emission estimation, and fair electricity cost allocation while considering occupants' economic positions? This is examined through the following sub-questions:

- 1) **RQ1:** How does electricity consumption differ between the two occupants based on room-level measurements?
- 2) **RQ2:** What are the corresponding carbon emissions from each occupant's electricity consumption?
- 3) **RQ3:** How does consumption-based electricity allocation differ from equal-bill allocation?
- 4) **RQ4:** How do differences in economic position relate to electricity cost burden?
- 5) **RQ5:** How can the developed web-based system support monitoring and visualization of electricity consumption, carbon emissions, and costs?

D. Limitations of the Study

This study is limited to two occupants and a 15-day monitoring period, so the findings cannot be generalized to other shared residential environments or economic groups. The economic-position comparison is limited to these occupants and indicates an association rather than a causal relationship with electricity consumption. The study also considers only electricity-related carbon emissions and does not include emissions from other household activities or sources.

II. METHODOLOGY

This study used a quantitative, IoT-based observational approach to measure room-level electricity consumption, carbon emissions, and electricity costs. Quantitative observation was appropriate because the study relied on numerical measurements collected systematically over time. Manual meter readings and questionnaire-based approaches were considered, but manual readings provide fewer observations and require repeated human intervention, while questionnaires depend on participants' ability to accurately report their electricity use. Automated IoT sub-metering was therefore selected because it provided direct, repeated measurements and was better suited to comparing electricity consumption between individual rooms.

A. Data Collection Methods

Electricity data were collected from Room F and Room K over a 15-day period using two Tuya-compatible smart meters. The meters communicated with the monitoring system through the local Wi-Fi network using the TinyTuya Python API, without relying on Tuya Cloud.

The monitoring system recorded measurements at a fixed interval of 10 seconds. For each successful reading, the following parameters were recorded:

- Voltage (V)
- Current (A)
- Instantaneous power (W)
- Cumulative energy consumption (kWh)
- Device status
- Switch status
- Fault status
- Timestamp

The readings were stored in a local SQLite database for analysis. Automated measurement was preferred because it provided actual high-frequency electricity data rather than relying on estimated or self-reported consumption. This approach is consistent with previous IoT energy-monitoring studies that used automated sensing to provide continuous electricity measurements [2], [6], [8]. However, the system was locally hosted and depended on a computer and network connection remaining available. Power outages or network interruptions could therefore interrupt data collection and require the system to be restarted.

The developed web-based system, built using Flask, Pandas, NumPy, SQLite, and TinyTuya, was used to monitor the devices and visualize the collected data. It provided live

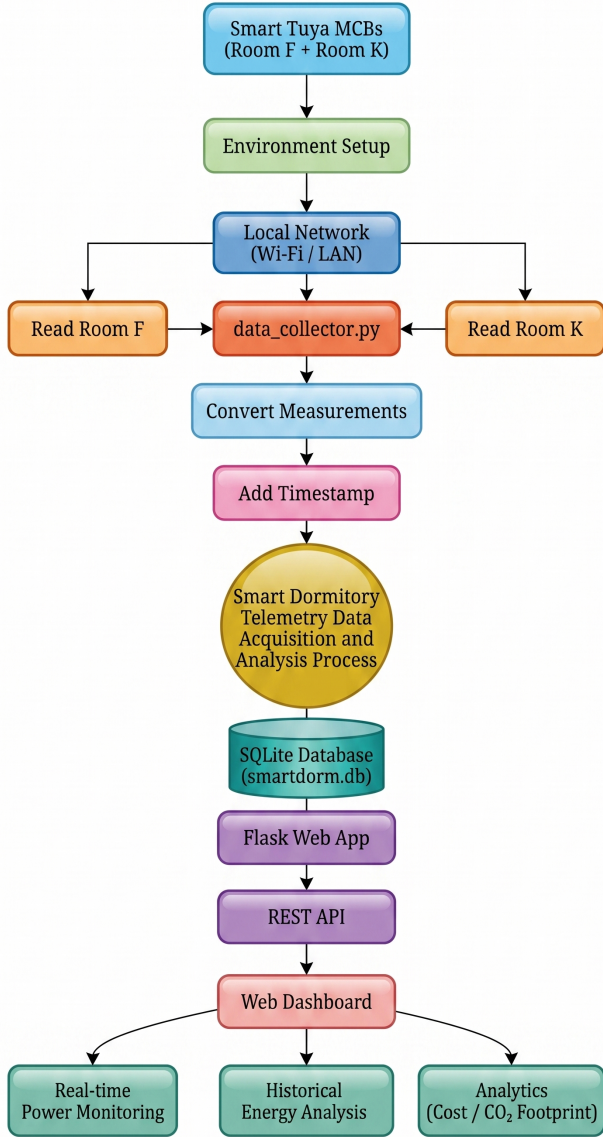


Fig. 1. Architecture of the proposed IoT-based smart sub-metering system.

monitoring, historical consumption trends, room comparisons, electricity-cost analysis, and carbon-emission estimates.

The instantaneous electrical power measured at each reading can be represented as:

$$P = V \times I \quad (1)$$

where P represents electrical power in watts, V represents voltage in volts, and I represents current in amperes.

B. Participants and Sampling

The study used purposive sampling to select two suitable rooms, Room F and Room K, occupied by two working individuals with different economic positions. The rooms were selected based on their suitability for installing and

operating the smart-metering system. Room K was occupied by a journalist, while Room F was occupied by a medicine company representative with a comparatively lower economic position.

Purposive sampling was appropriate because the study aimed to obtain detailed measurements from specific rooms rather than produce a statistically representative sample of all residential occupants. The resulting study was therefore treated as a small 15-day case study. This allowed detailed comparison between the two rooms but limited the generalizability of the findings to other dormitories and economic groups.

C. Data Analysis Method

The collected 15-day data were analyzed quantitatively using Python, Pandas, NumPy, and SQLite. The measurements were grouped by room and time period to examine energy consumption, average power, peak power, voltage, and electricity-use patterns. Descriptive aggregation was used to summarize the measured data and identify differences in consumption and power demand between the two rooms. The results were presented using graphs and the developed web-based dashboard.

For a measurement interval of Δt hours, the energy consumption associated with an individual reading can be estimated as:

$$E = P \times \Delta t \quad (2)$$

where E is energy consumption in kWh and P is power in kW.

Electricity costs were calculated using the measured energy consumption and an electricity rate of BDT 7.5/kWh. Each room's consumption-based cost was determined from its proportion of the total measured consumption and compared with an equal division of the total bill. Carbon emissions were estimated by multiplying each room's measured energy consumption by the Bangladesh grid emission factor of 0.637 kg CO₂/kWh, following:

$$C = E \times EF \quad (3)$$

where C represents carbon emissions, E represents electricity consumption in kWh, and EF represents the electricity-grid emission factor in kg CO₂e/kWh. These calculations provided the basis for comparing electricity consumption, power patterns, carbon emissions, and cost allocation.

D. Research Ethics

The study primarily collected technical electricity data rather than sensitive personal information. Room identifiers were used instead of personal names in the analysis. Access credentials, such as device keys, were kept private and were not included in the research outputs.

Because electricity patterns may indirectly reveal occupant behavior, the collected data were treated as research data and presented without unnecessary personally identifying information. The remote switching function was used as a system

TABLE I
SUMMARY OF ELECTRICITY CONSUMPTION OVER THE 15-DAY
OBSERVATION PERIOD

Room	Energy (kWh)	Peak Power (W)	Carbon (kg CO ₂ e)
Room F	40.61	≈ moderate	25.87
Room K	65.83	≈ 470	41.93

feature and operated with care to avoid unintended interruption of electricity supply.

E. Data Collection and Analysis

Electricity data were collected from Room F and Room K over a 15-day observation period using the developed IoT sub-metering system. Each smart meter was queried at 10-second intervals, and valid measurements were stored in a SQLite database. The collected variables included voltage, current, instantaneous power, cumulative energy consumption, device status, switch status, fault status, and timestamp.

The collected data were processed using Python, Pandas, and NumPy. Room-level measurements were aggregated to compare electricity consumption and power demand over time. Carbon emissions were calculated using the Bangladesh grid emission factor of 0.637 kg CO₂/kWh, while electricity costs were calculated using a rate of BDT 7.5/kWh. The analysis compared equal-bill and consumption-based allocation, with the results visualized through the developed web-based dashboard.

III. RESULTS

A. Data Collection and Analysis

Overall electricity consumption differed substantially between the two monitored rooms. During the 15-day observation period, Room K consumed 65.83 kWh, while Room F consumed 40.61 kWh, giving a difference of 25.22 kWh. Thus, Room K's measured electricity consumption was approximately 62% higher than Room F's. These differences indicate variations in appliance usage, occupancy behavior, and daily electricity demand between the two occupants.

The hourly power profile shows that the difference between the rooms was not uniform throughout the day. Room K generally exhibited higher power demand than Room F, with the largest differences occurring during the evening. On weekdays, Room K's average power increased sharply after approximately 17:00, reaching around 360 W at 23:00, while Room F remained comparatively low for most of the same period. During weekends and holidays, Room K also showed substantially higher demand, reaching approximately 470 W at 18:00. Room F showed a smaller increase during these periods, with its highest weekend/holiday values occurring around the evening.

These patterns indicate that the higher total consumption observed for Room K was accompanied by periods of substantially greater instantaneous power demand, particularly during evening and non-working-day periods.

(a) Total Energy Consumption & Carbon Emissions

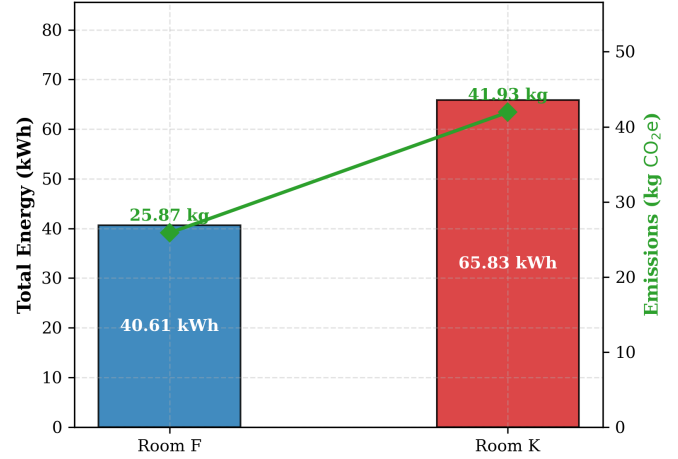


Fig. 2. Comparison of total electricity consumption and estimated carbon emissions for Room K and Room F.

B. Energy Consumption Analysis

The measured electricity consumption and corresponding environmental impact are summarized in Fig. 2. Room K recorded higher total electricity consumption than Room F throughout the observation period. The figure provides a comparative view of the total energy consumption and associated carbon emissions for the two rooms.

C. Carbon Footprint Estimation

The corresponding estimated carbon emissions followed the same pattern as measured energy consumption. Room K was associated with 41.93 kg CO₂, compared with 25.87 kg CO₂ for Room F. This represents an estimated difference of 16.06 kg CO₂ over the observation period, driven directly by the higher electricity consumption recorded for Room K.

The distribution of measured power demand is illustrated in Fig. 3. The results show that Room K experienced a wider and generally higher range of power demand compared with Room F.

D. Energy Consumption and Economic Position

Electricity consumption was compared across the two occupants according to their economic position. Room K, occupied by a journalist with a comparatively higher economic position, recorded both higher electricity consumption and a higher evening power demand than Room F, occupied by a medicine company representative with a comparatively lower economic position. This pattern is treated as an observed association within this two-occupant case study rather than evidence of a causal relationship.

E. Weekly and Holiday Consumption Pattern

The collected data were grouped according to the day of the week and categorized into weekday and weekend/holiday periods. Both rooms showed elevated power demand during

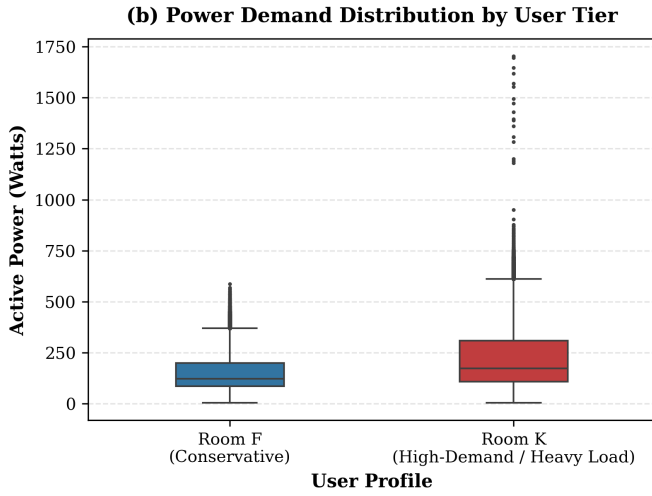


Fig. 3. Distribution of measured power demand for Room K and Room F during the observation period.

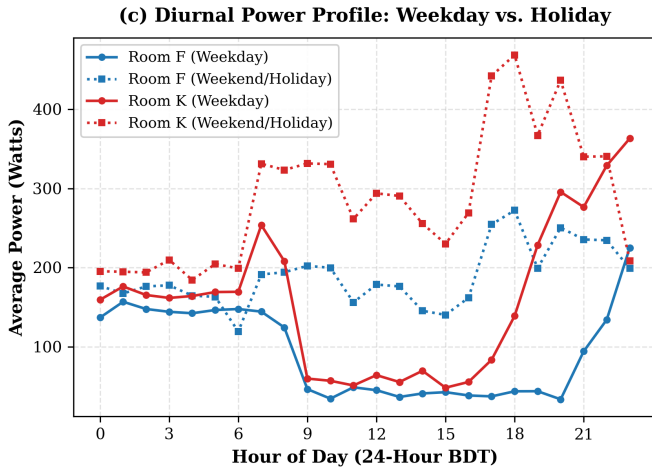


Fig. 4. Hourly power demand profiles for Room K and Room F during weekday and weekend/holiday periods.

weekend/holiday evenings relative to weekdays, but the increase was substantially larger for Room K, whose weekend/holiday peak power (≈ 470 W) exceeded its own weekday peak (≈ 360 W) as well as Room F's demand during the same periods.

The hourly variation in electricity demand across the observation period is presented in Fig. 4. The figure highlights the differences between weekday and weekend/holiday power profiles.

F. Fair Billing Analysis

In conventional shared-housing environments in Bangladesh, electricity costs are often divided equally among occupants, irrespective of differences in individual consumption. This flat-split billing approach can create cross-subsidization, where low-consumption occupants effectively

TABLE II
CONVENTIONAL AND IoT-BASED BILLING COMPARISON

Method	Room K	Room F	Total
50/50 Split	BDT 400.00	BDT 400.00	BDT 800.00
IoT Proportional	BDT 494.78	BDT 305.22	BDT 800.00
Difference	+BDT 94.78	-BDT 94.78	BDT 0.00

bear part of the electricity cost generated by high-consumption occupants.

Under conventional billing, the total electricity cost is divided equally among occupants:

$$Cost_i^{conv} = \frac{C_{total}}{N} \quad (4)$$

where C_{total} is the total electricity cost and N is the number of occupants.

In contrast, the proposed approach allocates costs according to measured electricity consumption:

$$Cost_i^{fair} = \frac{E_i}{E_{total}} C_{total} \quad (5)$$

where E_i is the electricity consumption of occupant i and E_{total} is the total measured consumption.

To quantify this billing inequity, a representative billing cycle with a total electricity cost of BDT 800.00 is considered. During the 14-day monitoring period, the two sub-metered units, Room K and Room F, recorded the energy consumption values reported above. The aggregate electricity consumption is therefore

$$E_{total} = E_K + E_F = 65.83 + 40.61 = 106.44 \text{ kWh}. \quad (6)$$

Applying the proportional-allocation model, the cost assigned to Room K is

$$C_K = \left(\frac{65.83}{106.44} \right) \times 800.00 \approx \text{BDT } 494.78, \quad (7)$$

and the cost assigned to Room F is

$$C_F = \left(\frac{40.61}{106.44} \right) \times 800.00 \approx \text{BDT } 305.22. \quad (8)$$

Under the conventional 50/50 billing method, each occupant would instead pay

$$C_{flat} = \frac{800.00}{2} = \text{BDT } 400.00. \quad (9)$$

The resulting differences between conventional and consumption-based billing are $\Delta C_K = 494.78 - 400.00 = +\text{BDT } 94.78$ for Room K and $\Delta C_F = 305.22 - 400.00 = -\text{BDT } 94.78$ for Room F, summarized in Table II.

The billing comparison is visually presented in Fig. 5. The consumption-based approach assigns a larger share of the total

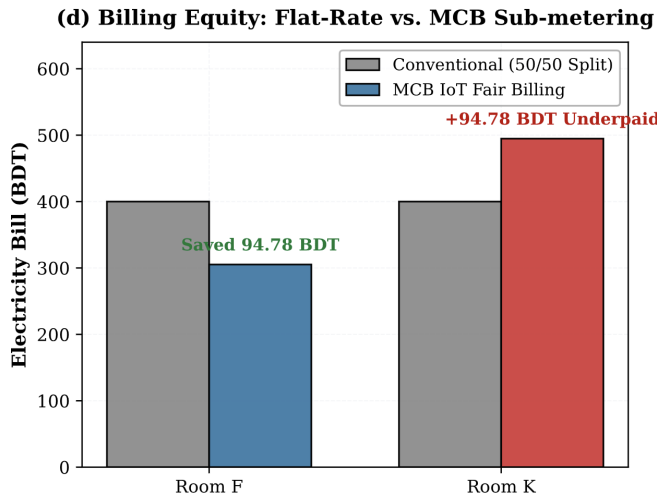


Fig. 5. Comparison of conventional equal-sharing and IoT-based proportional electricity billing for Room K and Room F.

cost to Room K because of its higher measured electricity consumption, while Room F receives a lower allocation.

The results demonstrate that the conventional flat-split method creates a cross-subsidy of BDT 94.78 from the lower-consuming occupant to the higher-consuming occupant. For Room F, this difference represents approximately

$$\frac{94.78}{400} \times 100 \approx 23.70\% \quad (10)$$

of its conventional bill. In contrast, IoT-enabled sub-metering aligns financial responsibility with actual measured consumption, thereby improving billing fairness, transparency, and accountability. ““

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